

Solution:

(a) *Lemma.* For any node u , the path $P(u) = \langle p_0, p_1, \dots, p_m \rangle$ produced by $\text{visit}(u)$ satisfies:

- (a) The last node p_m is one of the children of u .
- (b) The first edge (p_0, p_1) satisfies $d(u, p_1) \leq 2r_0$.
- (c) The closing edge (p_m, u) satisfies $d(p_m, u) \leq r_0$.

Proof. We prove by induction on the structure of T .

Base case: If u is a leaf, $P(u) = \langle u \rangle$, and all three properties hold trivially.

Inductive step: Suppose for each child u_i , the path $P(u_i)$ satisfies the lemma. Then:

- The last node p_m is the first node of $(P(u_k))^R$, which is u_k , a child of u .
- The first node after u is the last node of $(P(u_1))^R$, which is the last node of $P(u_1)$, say q_{m_1} . By inductive hypothesis, q_{m_1} is a descendant of u_1 . By the triangle inequality:

$$d(u, q_{m_1}) \leq d(u, u_1) + d(u_1, q_{m_1}) \leq r_0 + r_0 = 2r_0.$$

- The closing edge from p_m back to u connects u_k to u , and since u_k is a child of u , the edge has weight at most r_0 .

This completes the inductive proof of the lemma. □

Now consider the edges in the final cycle C :

- **Internal edges within each $(P(u_i))^R$:** These edges are either edges within a child subtree or its reverse. Since each subtree path corresponds to traversals in T , and every edge in T has weight at most r_0 , these internal edges are of weight at most r_0 .
- **Edges between subpaths:** When transitioning from the end of $(P(u_i))^R$ to the beginning of $(P(u_{i+1}))^R$, the two nodes involved are u_i (the child just completed) and some descendant $q_{m_{i+1}}$ of u_{i+1} . By the triangle inequality:

$$d(u_i, q_{m_{i+1}}) \leq d(u_i, u) + d(u, q_{m_{i+1}}).$$

Since u_i and u are connected by an edge in T of weight at most r_0 , and from the lemma $d(u, q_{m_{i+1}}) \leq 2r_0$, we get:

$$d(u_i, q_{m_{i+1}}) \leq r_0 + 2r_0 = 3r_0.$$

- **Closing edge:** Finally, the edge from p_m (which is u_k) back to u has weight at most r_0 by the lemma.

Thus, every edge in the constructed cycle C has weight at most $3r_0$.

(b) The algorithm provided in the question yields a polynomial-time, 3-approximation algorithm.

1. **Computing r_0 .** We sort all $\binom{n}{2}$ edge weights in non-decreasing order and scan them, maintaining the connectivity of the subgraph $G[r]$ using a Union-Find data structure. This allows us to determine the smallest r_0 such that $G[r_0]$ is connected. The time complexity of this step is $O(n^2 \log n)$.
2. **Building and traversing the spanning tree.** Once r_0 is known, we construct any spanning tree T of $G[r_0]$ using Prim's algorithm in $O(n^2)$ time. The recursive `visit` procedure then generates a path through all nodes in linear time relative to the number of vertices.
3. **Approximation.** As shown in (a), the final Hamiltonian cycle C has maximum edge weight at most $3r_0$. Since any valid Hamiltonian cycle in G must use only edges of weight $\leq r^*$, where r^* is the optimal value, and $r_0 \leq r^*$, we conclude:

$$\max_{e \in C} d(e) \leq 3r_0 \leq 3r^*.$$

Therefore, the algorithm runs in polynomial time and achieves a 3-approximation factor for the problem.